

Drinking Water Treatment: Teaching Synthesis

Evidence-based teaching resource

Evidence checked 2026-08-09

1 Currentness Notice

Current final drinking-water requirements reflect sources checked on 2026-08-09. Pending PFAS proposals are identified separately and do not establish final 2031 compliance timing. Check regulatory status before operational use.

2 Purpose and Scope

This chapter presents a concise, source-attributed teaching synthesis of Drinking Water Treatment. It distinguishes regulations, proposals, guidance, scientific assessments, and datasets.

3 Evidence and Authority Framework

A defensible application identifies the issuing organization, jurisdiction, effective status, relevant date, population or geographic scope, and stated limitations before using a claim.

4 Current Authoritative Baseline

World Health Organization WHO's fourth edition incorporating three addenda retains Water Safety Plans as a source-to-consumer risk-management approach.

U.S. Environmental Protection Agency EPA identifies national primary drinking-water regulations as legally enforceable standards for public water systems.

U.S. Environmental Protection Agency The current 2024 final rule sets the PFOA maximum contaminant level at 4.0 ng/L.

U.S. Environmental Protection Agency The current 2024 final rule sets the PFOS maximum contaminant level at 4.0 ng/L.

U.S. Environmental Protection Agency The current final rule uses a unitless Hazard Index MCL of 1 for specified PFAS mixtures.

U.S. Environmental Protection Agency The 2024 final rule’s current implementation summary requires initial PFAS monitoring by 2027 and treatment compliance by 2029 unless a later final action changes those dates.

U.S. Environmental Protection Agency EPA’s 2026 proposal would allow qualifying systems to request an extension of PFOA/PFOS compliance to April 26, 2031; it does not itself change the final deadline.

U.S. Environmental Protection Agency EPA proposed rescinding the individual/Hazard Index requirements for PFHxS, PFNA, HFPO-DA, and relevant mixtures; the proposal is not current final law.

U.S. Environmental Protection Agency EPA’s implementation tools are guidance supporting Lead and Copper Rule Improvements implementation; codified requirements remain controlling.

U.S. Environmental Protection Agency EPA lists treatment technologies including activated carbon, ion exchange, membrane processes, adsorptive media, biological treatment, and ultraviolet/advanced oxidation for specified applications.

U.S. Environmental Protection Agency SDWIS federal reporting data are submitted and frozen quarterly; submission year/quarter is version context rather than an event date.

U.S. Environmental Protection Agency ECHO’s SDWA downloads derive from SDWIS and their documentation must be used to interpret compliance fields and refresh timing.

5 Instructional Application

Learners classify source authority, test the boundaries of each claim, construct a hypothetical professional decision, and communicate what is known, proposed, recommended, or unresolved.

6 Common Errors

- Treating proposed regulatory text as current final law.
- Treating recommendations or guidance as automatically enforceable.
- Generalizing regional or dataset-specific findings beyond their documented scope.
- Confusing publication, release, effective, retrieval, and observation-period dates.

7 Limitations

This synthesis is not a legal opinion or an operational substitute for current agency text, site-specific professional judgment, or applicable state and local requirements.

8 Conclusion

Professional teaching is strengthened when factual claims are current, authority distinctions remain visible, uncertainty is explicit, and external sources are independently identifiable.

9 References

- World Health Organization. Guidelines for drinking-water quality: fourth edition incorporating the first, second and third addenda. <https://www.who.int/publications/i/item/9789240121225>
- World Health Organization. Drinking-water quality guidelines official landing page. <https://www.who.int/teams/environment-climate-change-and-health/water-sanitation-and-health/water-safety-and-quality/drinking-water-quality-guidelines>
- U.S. Environmental Protection Agency. National Primary Drinking Water Regulations. <https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-regulations>
- Electronic Code of Federal Regulations. 40 CFR Parts 141 and 142 current codified drinking-water requirements. <https://www.ecfr.gov/current/title-40/chapter-I/subchapter-D/part-141>
- U.S. Environmental Protection Agency. PFAS National Primary Drinking Water Regulation and implementation source. <https://www.epa.gov/sdwa/and-polyfluoroalkyl-substances-pfas>
- U.S. Environmental Protection Agency. Proposed PFOA and PFOS Compliance Extension Rule. <https://www.epa.gov/sdwa/proposed-pfoa-and-pfos-compliance-extension-rule>
- U.S. Environmental Protection Agency. Proposed PFAS Rescission Rule. <https://www.epa.gov/sdwa/proposed-pfas-rescission-rule>
- Office of the Federal Register. Official Federal Register record for proposed PFOA and PFOS compliance extension. <https://www.federalregister.gov/d/2026-10086>
- U.S. Environmental Protection Agency. Lead and Copper Rule implementation tools. <https://www.epa.gov/dwreginfo/lead-and-copper-rule-implementation-tools>
- U.S. Environmental Protection Agency. Overview of Drinking Water Treatment Technologies. <https://www.epa.gov/sdwa/overview-drinking-water-treatment-technologies>
- U.S. Environmental Protection Agency. SDWIS Federal Reporting Services Data Dictionary. <https://www.epa.gov/DWdata/sdwis-federal-reporting-services-data-dictionary>
- U.S. Environmental Protection Agency. ECHO Safe Drinking Water Act download documentation. <https://echo.epa.gov/tools/data-downloads/sdwa-download-summary>